Mahmoud Alyosify

Software Engineer

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SUMMARY

Software engineer with production experience on **Egypt's national toll systems** (C#, .NET, SQL Server) and current work building tested Python services with FastAPI. Full-stack trained in .NET and Angular, with a strong base in data structures, algorithms and databases. MSc in Artificial Intelligence, Queen's University, Canada (GPA 4.30 / 4.30), useful for teams adding AI to their products.

SKILLS

Backend: C#, .NET, ASP.NET Core, Entity Framework, Python, FastAPI, Pydantic, REST APIs
Frontend: Angular, TypeScript, JavaScript, HTML, CSS
Databases: SQL Server, MySQL, MongoDB, Crystal Reports
Practices: automated testing (pytest), code review, debugging production systems, Git, Docker, Linux
AI: LLMs, RAG, LangChain, machine learning

EXPERIENCE

AI Engineer Intern — Oct 2026 – Present
Injaz Tech (Injaz Group, formerly Fingerprint Consultancy) — Maadi, Cairo

- Built the **FastAPI** service: typed request and response models (Pydantic), streaming answers, session memory, health checks and rate limiting.
- Covered the service with **136 automated tests** (pytest) and an evaluation check that compares each new version with the last.
- Wrote the **Docker** setup and a **GitHub Actions** workflow (linting, 136 tests, evaluation gate); added health checks, Prometheus metrics, retries and a circuit breaker.
- Building a RAG assistant that answers employees' questions from company documents, using **LangChain, LangGraph, FAISS** and **FastAPI**.

Systems Engineer (Military Service) — Feb 2024 – Mar 2025
National Company for Roads Building & Development — Cairo

- Supported the backend systems of Egypt's national toll network, used by **more than half of the country's vehicles** across 70%+ of national roads.
- Co-developed an internal data-management and reporting application in **C#, .NET, Entity Framework, SQL Server** and Crystal Reports.
- Debugged live transportation software and supported legacy applications to keep downtime to a minimum.

Part-Time Machine Learning Instructor — Jul 2025 – Present
National Telecommunication Institute (NTI) — Egypt

- Teach **90-hour machine learning bootcamps** and AI for Business modules in NTI's Summer Internships Program.

Online Instructor — 2020 – Present
Udemy & YouTube (Arabic courses) — Remote

- Created Arabic courses on algorithms, discrete mathematics, probability and statistics: **10,000+ learners, 700K+ views**.

PROJECTS

Inventory Management System
Desktop application, SQL database

- Inventory system with barcode scanning and low-stock alerts, from database design to the user workflow.

E-Commerce Web Application
JavaScript, JWT, Stripe

- Shopping front end with secure JWT login and Stripe payments.

Mini-FAT File System
C

- A FAT-style file system written from first principles: allocation table, directories and file operations.

Human vs. AI-Agent Pull Requests — research study
Python, GitHub data mining, statistics

- Built a reproducible pipeline that mines GitHub to compare review effort on pull requests written by AI agents and by people.

Automated Multimodal Agent — PDF to narrated slides 2026
Python, LangGraph, Hugging Face, FAISS, Streamlit, text-to-speech

- An agent that turns a PDF into a narrated PowerPoint deck: it reads the document, plans and writes the slides, and records the narration.

Also: MinimalLM (Deloitte-mentored MSc thesis, 47.6% fewer LLM tokens) · HORUS Sentinel (multi-agent OSINT) · SimCLR-Vision-SSL (84.30% top-1, self-supervised).

PUBLICATIONS

Say Less, Mean Better: Training-Free Verbosity Control for Language Models
M. Alyosify, M. Imbabi, E. Elkhamisi, R. Ferdousi · arXiv:2611.04107 (cs.CL) · November 2026

Beyond Truncation? Entailment-Licensed Elision and the Limits of Semantic Preservation in LLM Decoding
M. Alyosify, M. Imbabi, E. Elkhamisi · arXiv:2611.08975 (cs.CL) · November 2026

EDUCATION

MSc in Artificial Intelligence Sep 2025 – Oct 2026
Queen's University Kingston, Canada

- GPA **4.30 / 4.30** · Completed October 2026 · **Digilians Presidential Scholarship** (competitive national AI fellowship)
- Coursework: Large Language Models, Generative AI, Deep Learning, Reinforcement Learning, Cloud Computing, End-to-End ML Pipelines

BSc in Computer & Information Science (Bioinformatics) Sep 2019 – Jul 2023
Assiut University Assiut, Egypt

- GPA 3.53 / 4.00 (Very Good with Honours) · Graduation project graded A+
- Strong base in data structures, algorithms, operating systems and databases; about **100 competitive-programming problems** solved on Codeforces
- Graduation project: Vitalism Solution, vital signs from ordinary video; **Microsoft Imagine Cup 2023 semi-finalist**

CERTIFICATIONS

- Full-Stack Development Diploma, .NET & Angular (Route IT Training Center, 2023–2024)
- ITI courses: .NET web development, SQL Server & C#, MEARN stack, Android (300+ hours)
- IBM AI Developer Professional Certificate (IBM, 2026)
- AWS Certified Cloud Practitioner (Amazon Web Services)

AWARDS & LEADERSHIP

- 1st Place, Smart Cities Hackathon 2022 (77 university teams)
- Semi-Finalist, Microsoft Imagine Cup 2023
- Student Ambassador, Information Technology Institute (ITI)

LANGUAGES

Arabic (native) · English (professional working proficiency)